

B6 Condensed-Matter Physics — Model Solutions, 2020

Question 1 — FCC structure factor and Si powder diffraction

Problem. (a) Derive the FCC selection rule and show the per-basis-atom contribution is $4f e^{2\pi i(hx+ky+lz)}$. (b) Show $|S_{\{hkl\}}|^2 = 32f_{\text{Si}}^2(1 + \cos(n\pi/2))$ for diamond Si and identify n . (c) Give d_{hkl} and powder multiplicities for (002), (111). (d) Complete a powder-pattern table for neutrons at $\lambda = 0.1542 \text{ nm}$. (e) X-ray vs. neutron intensity ratio as a function of 2θ .

(a) FCC extinction rule

Four FCC atoms at $\mathbf{r} + (0, 0, 0)$, $\mathbf{r} + (\frac{1}{2}, \frac{1}{2}, 0)$, $\mathbf{r} + (\frac{1}{2}, 0, \frac{1}{2})$, $\mathbf{r} + (0, \frac{1}{2}, \frac{1}{2})$ with $\mathbf{r} = (x, y, z)$. Sum the plane-wave phases:

$$S_{\text{FCC}}(hkl) = f e^{2\pi i(hx+ky+lz)} [1 + e^{i\pi(h+k)} + e^{i\pi(h+l)} + e^{i\pi(k+l)}].$$

Reduce each exponential using $e^{i\pi m} = (-1)^m$:

$$e^{i\pi(h+k)} = (-1)^{h+k}, \quad e^{i\pi(h+l)} = (-1)^{h+l}, \quad e^{i\pi(k+l)} = (-1)^{k+l}.$$

All h, k, l same parity (all even / all odd) makes every index sum even:

$$1 + 1 + 1 + 1 = 4.$$

Mixed parity: exactly two of the three sums $(h+k, h+l, k+l)$ are odd:

$$1 + 1 - 1 - 1 = 0.$$

Therefore the bracket equals 4 when h, k, l are all even or all odd, and 0 otherwise:

$$S_{\text{FCC}}(hkl) = 4f e^{2\pi i(hx+ky+lz)} \quad (\text{non-extinct: } h, k, l \text{ all even or all odd}). \quad (1)$$

(b) Diamond-Si structure factor

Two basis atoms at $(0, 0, 0)$ and $(\frac{3}{4}, \frac{1}{4}, \frac{3}{4})$, each carrying the FCC factor (1):

$$S_{\{hkl\}} = 4f_{\text{Si}} [1 + e^{2\pi i(3h/4+k/4+3l/4)}].$$

Collect the exponent over a common denominator 4:

$$2\pi\left(\frac{3h}{4} + \frac{k}{4} + \frac{3l}{4}\right) = \frac{2\pi(3h+k+3l)}{4} = \frac{\pi}{2}(3h+k+3l).$$

Define $n = 3h + k + 3l$ (integer):

$$S_{\{hkl\}} = 4f_{\text{Si}} [1 + e^{in\pi/2}].$$

Take modulus squared with $|1 + e^{i\phi}|^2 = 2(1 + \cos \phi)$:

$$|S_{\{hkl\}}|^2 = 16f_{\text{Si}}^2 \cdot 2\left(1 + \cos \frac{n\pi}{2}\right).$$

Combine the constants:

$$|S_{\{hkl\}}|^2 = 32f_{\text{Si}}^2 \left(1 + \cos \frac{n\pi}{2}\right), \quad n = 3h + k + 3l. \quad (2)$$

(c) Plane spacing and multiplicities

Cubic lattice:

$$d_{(hkl)} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}.$$

$m_{(002)}$: $\{200\}$ -type permutations with one non-zero index ± 2 :

$$(\pm 2, 0, 0), (0, \pm 2, 0), (0, 0, \pm 2) \implies m_{(002)} = 6.$$

$m_{(111)}$: all sign combinations of $(1, 1, 1)$:

$$2 \times 2 \times 2 = 8 \implies m_{(111)} = 8.$$

(d) Neutron powder table

Bragg's law: $\sin \theta = \lambda / (2d_{hkl})$, $\lambda = 0.1542 \text{ nm}$, $a = 0.543 \text{ nm}$.

For each peak compute $n = 3h + k + 3l$, $|S|^2 / f_{\text{Si}}^2$ from (2), multiplicity m :

- (111): $n = 7$, $\cos(7\pi/2) = 0$, $|S|^2 = 32f^2$; $m = 8$.
- (200): $n = 6$, $\cos(3\pi) = -1$, $|S|^2 = 0$; $m = 6$ (extinct).
- (220): $n = 8$, $\cos(4\pi) = 1$, $|S|^2 = 64f^2$; $m = 12$.
- (311): $n = 13$, $\cos(13\pi/2) = 0$, $|S|^2 = 32f^2$; $m = 24$.
- (400): $n = 12$, $\cos(6\pi) = 1$, $|S|^2 = 64f^2$; $m = 6$.

Compute $d_{hkl} = a / \sqrt{h^2 + k^2 + l^2}$:

$$d_{111} = \frac{0.543}{\sqrt{3}} = 0.3135 \text{ nm}, \quad d_{200} = \frac{0.543}{2} = 0.2715 \text{ nm}, \quad d_{220} = \frac{0.543}{\sqrt{8}} = 0.1920 \text{ nm}.$$

$$d_{311} = \frac{0.543}{\sqrt{11}} = 0.1637 \text{ nm}, \quad d_{400} = \frac{0.543}{4} = 0.1358 \text{ nm}.$$

Then $2\theta = 2 \arcsin[\lambda / (2d)]$. For (111):

$$\sin \theta_{111} = \frac{0.1542}{2 \cdot 0.3135} = 0.2460 \implies 2\theta_{111} = 2 \arcsin(0.2460) = 28.47^\circ.$$

The remaining peaks follow identically; tabulated below:

peak	$\{hkl\}$	$d_{\{hkl\}}/\text{nm}$	$2\theta_{\{hkl\}}/^\circ$	m	$I_{\text{calc}}/f_{\text{Si}}^2$	I_{meas}
a	111	0.3135	28.47	8	256	1000
b	200	0.2715	33.01	6	0	0
c	220	0.1920	47.40	12	768	2755
d	311	0.1637	56.18	24	768	2619
e	400	0.1358	69.20	6	384	1203

Ratio $I_{\text{meas}}/I_{\text{calc}}$ (should be a single constant $\propto b_{\text{Si}}^2$):

$$(a) 3.91, \quad (c) 3.59, \quad (d) 3.41, \quad (e) 3.13.$$

Ratios decrease with 2θ : **Debye–Waller damping** e^{-2W} with $W \propto \sin^2 \theta / \lambda^2$ suppresses high-angle peaks. Otherwise the agreement (constancy of the ratio) is good.

(e) X-rays vs. neutrons

(i) Replace the neutron Fermi length b_{Si} by the X-ray atomic form factor $f_{\text{Si}}(Q)$, which is Q -dependent (Fourier transform of the electron density), so $|S|^2 = 32 f_{\text{Si}}^2(Q) (1 + \cos(n\pi/2))$.

(ii) $f_{\text{Si}}(Q)$ falls monotonically with Q (and thus with 2θ), while b_{Si} is Q -independent (point nuclear scatterer). Hence

$$\frac{I_X}{I_n} \propto \frac{f_{\text{Si}}^2(2\theta)}{b_{\text{Si}}^2} \quad \text{decreases monotonically with } 2\theta_{hkl}.$$

Question 2 — 2D Debye theory and graphene

Problem. (a) Derive 2D Debye wavevector. (b) Find ω_D , T_D , $g(\omega)$ for linear ($\omega = ck$, two polarisations) and quadratic ($\omega = Dk^2$, one polarisation) modes. (c) Low- T molar specific heat in form $R\alpha(T/T_D)^\beta$. (d) Apply to graphene: $c_v = R(c_1T + c_2T^2)$, find c_1, c_2 . (e) Compare to room- T data.

(a) 2D Debye wavevector

One atom per cell in 2D has $2N$ vibrational degrees of freedom, hence N modes per polarisation. Equate the k -space disc area times the 2D mode density to the mode count:

$$\frac{A}{(2\pi)^2} \pi k_D^2 = N.$$

Rearrange for k_D^2 :

$$k_D^2 = \frac{4\pi^2 N}{\pi A} = \frac{4\pi N}{A}.$$

Use $n = N/A$:

$$k_D^2 = 4\pi n.$$

Take square root:

$$k_D = 2\sqrt{\pi n}, \quad a = 2\sqrt{\pi}. \quad (3)$$

(b) ω_D , T_D , $g(\omega)$

(i) Linear mode $\omega = ck$, two polarisations.

Cut-off:

$$\omega_D = ck_D = 2c\sqrt{\pi n}.$$

Debye temperature:

$$k_B T_D = \hbar \omega_D = 2\hbar c \sqrt{\pi n}.$$

DOS per polarisation per unit area from a 2D shell of radius k :

$$g_1(\omega) d\omega = \frac{1}{(2\pi)^2} \cdot 2\pi k dk = \frac{k}{2\pi} dk.$$

Substitute $k = \omega/c$, $dk = d\omega/c$:

$$g_1(\omega) d\omega = \frac{1}{2\pi} \cdot \frac{\omega}{c} \cdot \frac{d\omega}{c}.$$

Divide both sides by $d\omega$:

$$g_1(\omega) = \frac{\omega}{2\pi c^2}.$$

Multiply by two degenerate polarisations:

$$g(\omega) = 2g_1(\omega) = \frac{\omega}{\pi c^2}.$$

Read off:

$$\boxed{g(\omega) = \gamma \omega, \quad \gamma = \frac{1}{\pi c^2} \quad (\text{per unit area}).} \quad (4)$$

(ii) Quadratic mode $\omega = Dk^2$, one polarisation.

Cut-off:

$$\omega_D = D k_D^2 = 4\pi D n.$$

Debye temperature:

$$\boxed{k_B T_D = \hbar \omega_D = 4\pi \hbar D n.}$$

DOS per unit area: invert $\omega = Dk^2$:

$$k = \sqrt{\omega/D}, \quad dk = \frac{d\omega}{2\sqrt{D\omega}}.$$

Single polarisation contribution:

$$g(\omega) d\omega = \frac{k}{2\pi} dk = \frac{1}{2\pi} \sqrt{\frac{\omega}{D}} \cdot \frac{d\omega}{2\sqrt{D\omega}}.$$

Cancel $\sqrt{\omega}$ in numerator and denominator:

$$g(\omega) d\omega = \frac{1}{4\pi D} d\omega.$$

Divide both sides by $d\omega$:

$$\boxed{g(\omega) = \gamma', \quad \gamma' = \frac{1}{4\pi D} \quad (\text{per unit area}).} \quad (5)$$

(c) Low- T specific heat

Energy per unit area at temperature T :

$$u(T) = \int_0^{\omega_D} g(\omega) \frac{\hbar \omega}{e^{\beta \hbar \omega} - 1} d\omega.$$

Low- T limit: extend upper limit to ∞ .

Case (i): substitute (4):

$$u = \int_0^\infty \frac{1}{\pi c^2} \omega \frac{\hbar \omega}{e^{\beta \hbar \omega} - 1} d\omega = \frac{\hbar}{\pi c^2} \int_0^\infty \frac{\omega^2}{e^{\beta \hbar \omega} - 1} d\omega.$$

Substitute $x = \beta \hbar \omega$, $\omega = x/(\beta \hbar)$, $d\omega = dx/(\beta \hbar)$:

$$u = \frac{\hbar}{\pi c^2} \cdot \frac{1}{(\beta \hbar)^3} \int_0^\infty \frac{x^2 dx}{e^x - 1}.$$

Apply $\int_0^\infty x^2/(e^x - 1) dx = 2\zeta(3)$:

$$u = \frac{2\zeta(3) \hbar}{\pi c^2 (\beta \hbar)^3}.$$

Substitute $\beta\hbar = \hbar/(k_B T)$, so $(\beta\hbar)^3 = \hbar^3/(k_B T)^3$:

$$u = \frac{2\zeta(3) \hbar (k_B T)^3}{\pi c^2 \hbar^3} = \frac{2\zeta(3) (k_B T)^3}{\pi c^2 \hbar^2}.$$

Differentiate u with respect to T :

$$c_v^{\text{area}} = \frac{\partial u}{\partial T} = \frac{6\zeta(3) k_B^3 T^2}{\pi c^2 \hbar^2}.$$

Per atom divide by n ; molar multiply by N_A and use $R = N_A k_B$:

$$c_v = \frac{N_A}{n} c_v^{\text{area}} = R \cdot \frac{6\zeta(3) k_B^2 T^2}{n \pi c^2 \hbar^2}.$$

From (b)(i), $k_B T_D = 2\hbar c \sqrt{\pi n}$. Square both sides:

$$(k_B T_D)^2 = 4\pi n c^2 \hbar^2.$$

Invert to extract the prefactor:

$$\frac{1}{n \pi c^2 \hbar^2} = \frac{4}{(k_B T_D)^2}.$$

Multiply by $k_B^2 T^2$:

$$\frac{k_B^2 T^2}{n \pi c^2 \hbar^2} = \frac{4 T^2}{T_D^2}.$$

Substitute:

$$c_v = 24 \zeta(3) R \left(\frac{T}{T_D} \right)^2, \quad \alpha = 24\zeta(3), \quad \beta = 2.$$

Case (ii): substitute (5):

$$u = \frac{1}{4\pi D} \int_0^\infty \frac{\hbar \omega \, d\omega}{e^{\beta \hbar \omega} - 1}.$$

Substitute $x = \beta \hbar \omega$, $\omega = x/(\beta \hbar)$, $d\omega = dx/(\beta \hbar)$:

$$u = \frac{1}{4\pi D} \cdot \frac{\hbar}{(\beta \hbar)^2} \int_0^\infty \frac{x \, dx}{e^x - 1}.$$

Apply $\int_0^\infty x/(e^x - 1) \, dx = \pi^2/6$:

$$u = \frac{\hbar}{4\pi D (\beta \hbar)^2} \cdot \frac{\pi^2}{6}.$$

Cancel one $\hbar$ and combine constants:

$$u = \frac{\pi}{24 D \hbar \beta^2}.$$

Substitute $1/\beta = k_B T$:

$$u = \frac{\pi (k_B T)^2}{24 D \hbar}.$$

Differentiate with respect to T :

$$c_v^{\text{area}} = \frac{\partial u}{\partial T} = \frac{\pi k_B^2 \cdot 2T}{24 D \hbar} = \frac{\pi k_B^2 T}{12 D \hbar}.$$

Per mole, multiply by N_A/n and use $N_A k_B = R$:

$$c_v = \frac{N_A}{n} c_v^{\text{area}} = R \cdot \frac{\pi k_B T}{12 D \hbar n}.$$

From (b)(ii), $k_B T_D = 4\pi \hbar D n$, so $D \hbar n = k_B T_D / (4\pi)$:

$$\frac{1}{D \hbar n} = \frac{4\pi}{k_B T_D}.$$

Substitute:

$$c_v = \frac{R \pi k_B T}{12} \cdot \frac{4\pi}{k_B T_D} = \frac{4\pi^2 R k_B T}{12 k_B T_D} = \frac{\pi^2 R}{3} \cdot \frac{T}{T_D}.$$

Box:

$$c_v = \frac{\pi^2}{3} R \left(\frac{T}{T_D} \right), \quad \alpha = \pi^2/3, \quad \beta = 1.$$

(d) Graphene low- T specific heat

Areal number density: $n = \sigma / m_C$ with $m_C = M_C / N_A$. Mass per C atom:

$$m_C = \frac{12.01 \times 10^{-3} \text{ kg mol}^{-1}}{6.022 \times 10^{23} \text{ mol}^{-1}} = 1.994 \times 10^{-26} \text{ kg}.$$

Areal density (note $\sigma = 0.38 \times 10^{-3} \text{ g m}^{-2} = 0.38 \times 10^{-6} \text{ kg m}^{-2}$):

$$n = \frac{0.38 \times 10^{-6}}{1.994 \times 10^{-26}} = 1.906 \times 10^{19} \text{ m}^{-2}.$$

In-plane (linear) Debye temperature:

$$k_B T_D^{(1)} = 2 \hbar c \sqrt{\pi n}.$$

Evaluate πn :

$$\pi n = 3.1416 \cdot 1.906 \times 10^{19} = 5.989 \times 10^{19} \text{ m}^{-2}.$$

Take square root:

$$\sqrt{\pi n} = 7.739 \times 10^9 \text{ m}^{-1}.$$

Multiply through with $\hbar = 1.0546 \times 10^{-34} \text{ J s}$, $c = 16\,800 \text{ m/s}$:

$$k_B T_D^{(1)} = 2 \cdot 1.0546 \times 10^{-34} \cdot 16800 \cdot 7.739 \times 10^9.$$

Group powers of ten:

$$k_B T_D^{(1)} = 2 \cdot 1.0546 \cdot 16800 \cdot 7.739 \times 10^{-25}.$$

Reduce the numerical factor:

$$k_B T_D^{(1)} = 2.742 \times 10^{-20} \text{ J}.$$

Divide by k_B :

$$T_D^{(1)} = \frac{2.742 \times 10^{-20}}{1.381 \times 10^{-23}} = 1985 \text{ K}.$$

Out-of-plane (quadratic) Debye temperature:

$$k_B T_D^{(2)} = 4\pi \hbar D n.$$

Evaluate the prefactor:

$$4\pi Dn = 4\pi \cdot 6.06 \times 10^{-7} \cdot 1.906 \times 10^{19}.$$

Reduce:

$$4\pi Dn = 1.452 \times 10^{14} \text{ s}^{-1}.$$

Multiply by $\hbar$:

$$k_B T_D^{(2)} = 1.0546 \times 10^{-34} \cdot 1.452 \times 10^{14} = 1.531 \times 10^{-20} \text{ J}.$$

Divide by k_B :

$$T_D^{(2)} = \frac{1.531 \times 10^{-20}}{1.381 \times 10^{-23}} = 1109 \text{ K}.$$

Sum the two contributions (one mole of atoms):

$$c_v = \frac{\pi^2}{3} R \frac{T}{T_D^{(2)}} + 24\zeta(3) R \left(\frac{T}{T_D^{(1)}} \right)^2 = R(c_1 T + c_2 T^2).$$

Read off c_1 :

$$c_1 = \frac{\pi^2}{3T_D^{(2)}} = \frac{9.870}{3 \cdot 1109}.$$

Reduce:

$$c_1 = \frac{9.870}{3327} = 2.97 \times 10^{-3} \text{ K}^{-1}.$$

Read off c_2 :

$$c_2 = \frac{24\zeta(3)}{(T_D^{(1)})^2} = \frac{24 \cdot 1.202}{1985^2}.$$

Reduce:

$$c_2 = \frac{28.85}{3.940 \times 10^6} = 7.32 \times 10^{-6} \text{ K}^{-2}.$$

Box:

$$c_v = R(c_1 T + c_2 T^2), \quad c_1 \approx 2.97 \times 10^{-3} \text{ K}^{-1}, \quad c_2 \approx 7.32 \times 10^{-6} \text{ K}^{-2}.$$

(e) Room-temperature comparison

Insert $T = 300 \text{ K}$ into the linear term:

$$c_1 T = 2.97 \times 10^{-3} \cdot 300 = 0.891.$$

Insert into the quadratic term:

$$c_2 T^2 = 7.32 \times 10^{-6} \cdot 9 \times 10^4 = 0.659.$$

Sum:

$$c_1 T + c_2 T^2 = 0.891 + 0.659 = 1.550.$$

Multiply by R :

$$c_v(300 \text{ K}) = 8.314 \cdot 1.550 = 12.9 \text{ J K}^{-1} \text{ mol}^{-1}.$$

Experiment: $9 \text{ J K}^{-1} \text{ mol}^{-1}$. The low- T approximation *is not justified* at 300 K : $T/T_D^{(2)} = 0.27$, so the quadratic-mode integral has begun to saturate (each branch is bounded by R), and the linear extrapolation overshoots. A full integration up to ω_D would cap each branch at R and reduce the total.

Two atoms per unit cell: it doubles the number of branches (acoustic + optical), but the optical phonon energies are large ($\hbar\omega_{\text{opt}} \gtrsim k_B \cdot 1000 \text{ K}$ in graphene), so they remain frozen out at 300 K and contribute negligibly. The atom count per mole is unchanged, so c_v per mole of atoms is essentially unaffected by the basis.

Question 3 — Semiconductor statistics and GaAs

Problem. (a) Define symbols, justify parabolic bands and Boltzmann statistics. (b) Heavy/light hole ratio. (c) Effective hole mass m_p^* . (d) Law of mass action; carrier densities for undoped and p -doped GaAs at 300 K and 150 K. (e) Hall measurement.

(a) Definitions and approximations

(i) μ = chemical potential (Fermi level); m_e^* = conduction-band effective mass (curvature at the minimum); m_h^* = valence-band effective mass (curvature at the maximum); ϵ_c = conduction-band edge; ϵ_v = valence-band edge. Energy gap:

$$E_g = \epsilon_c - \epsilon_v.$$

(ii) Parabolic-band approximation: near each band edge $\epsilon(\mathbf{k}) \approx \epsilon_{c,v} \pm \hbar^2 k^2 / (2m^*)$. This gives the 3D DOS $g(\epsilon) \propto \sqrt{|\epsilon - \epsilon_{c,v}|}$, leading to the $(2m^*/(\beta\pi\hbar^2))^{3/2}$ prefactors. Validity: thermally excited carriers occupy states within $\sim k_B T$ of the edge, where the dispersion is well represented by its leading-order Taylor expansion.

(iii) For a non-degenerate gap $\mu \simeq \epsilon_v + E_g/2$, so $\epsilon_c - \mu \simeq E_g/2 = 0.75 \text{ eV}$. At 300 K $k_B T = 0.0259 \text{ eV}$:

$$\beta(\epsilon_c - \mu) \simeq 0.75/0.0259 \approx 29 \gg 1.$$

Hence $e^{\beta(\epsilon - \mu)} \gg 1$ for every conduction state and Fermi–Dirac collapses to Boltzmann.

(b) Heavy-to-light hole ratio

Both bands share ϵ_v and the same Boltzmann factor $e^{-\beta(\mu - \epsilon_v)}$; they differ only through $m^{*3/2}$:

$$\frac{p_{\text{hh}}}{p_{\text{lh}}} = \frac{(m_{\text{hh}}^*)^{3/2}}{(m_{\text{lh}}^*)^{3/2}} = \left(\frac{m_{\text{hh}}^*}{m_{\text{lh}}^*} \right)^{3/2}.$$

Insert numbers:

$$\frac{p_{\text{hh}}}{p_{\text{lh}}} = \left(\frac{0.45}{0.082} \right)^{3/2}.$$

Divide:

$$\frac{0.45}{0.082} = 5.488.$$

Take the square root:

$$\sqrt{5.488} = 2.343.$$

Take the $3/2$ power:

$$(5.488)^{3/2} = 5.488 \cdot 2.343 = 12.86.$$

Box:

$$\frac{p_{\text{hh}}}{p_{\text{lh}}} = 12.86,$$

which is independent of T since the Boltzmann exponentials cancel.

(c) Effective hole mass

Add the two valence-band populations:

$$p_{\text{tot}} = p_{\text{lh}} + p_{\text{hh}} = \frac{1}{4} \left(\frac{2}{\beta \pi \hbar^2} \right)^{3/2} e^{-\beta(\mu - \epsilon_v)} (m_{\text{lh}}^{*3/2} + m_{\text{hh}}^{*3/2}).$$

Identify with the target form $p_{\text{tot}} = \frac{1}{4} (2m_p^*/(\beta \pi \hbar^2))^{3/2} e^{-\beta(\mu - \epsilon_v)}$:

$$(m_p^*)^{3/2} = m_{\text{lh}}^{*3/2} + m_{\text{hh}}^{*3/2}.$$

Raise both sides to $2/3$:

$$m_p^* = (m_{\text{lh}}^{*3/2} + m_{\text{hh}}^{*3/2})^{2/3}.$$

Numerical value:

$$(0.082)^{3/2} = 0.082 \cdot \sqrt{0.082} = 0.082 \cdot 0.2864 = 0.0235.$$

$$(0.45)^{3/2} = 0.45 \cdot \sqrt{0.45} = 0.45 \cdot 0.6708 = 0.3018.$$

Sum:

$$m_p^{*3/2}/m_0^{3/2} = 0.0235 + 0.3018 = 0.3253.$$

Raise to $2/3$:

$$m_p^*/m_0 = (0.3253)^{2/3} = 0.471.$$

(d) Law of mass action and carrier densities

Multiply $n(T) p_{\text{tot}}(T)$; μ cancels because $(\epsilon_c - \mu) + (\mu - \epsilon_v) = E_g$:

$$np = \frac{1}{16} \left(\frac{2}{\beta \pi \hbar^2} \right)^3 (m_e^* m_p^*)^{3/2} e^{-\beta E_g} \equiv n_i^2. \quad (6)$$

Convenient form: at 300 K the effective DOS at $m^* = m_0$ is $N_0 \equiv \frac{1}{4} (2m_0 k_B T / (\pi \hbar^2))^{3/2}$.

Compute $k_B T$ at 300 K:

$$k_B T = 1.381 \times 10^{-23} \cdot 300 = 4.143 \times 10^{-21} \text{ J}.$$

Compute $2m_0 k_B T / (\pi \hbar^2)$:

$$\frac{2 \cdot 9.109 \times 10^{-31} \cdot 4.143 \times 10^{-21}}{\pi \cdot (1.0546 \times 10^{-34})^2}.$$

Reduce numerator and denominator:

$$\frac{7.547 \times 10^{-51}}{3.494 \times 10^{-68}} = 2.160 \times 10^{17} \text{ m}^{-2}.$$

Raise to the $3/2$:

$$(2.160 \times 10^{17})^{3/2} = 2.160 \times 10^{17} \cdot \sqrt{2.160 \times 10^{17}} = 1.004 \times 10^{26} \text{ m}^{-3}.$$

Divide by 4:

$$N_0(300 \text{ K}) = \frac{1}{4} \cdot 1.004 \times 10^{26} = 2.51 \times 10^{25} \text{ m}^{-3}.$$

Therefore at 300 K:

$$n_i^2 = N_0^2 (m_e^* m_p^* / m_0^2)^{3/2} e^{-\beta E_g}.$$

Insert masses:

$$0.067 \cdot 0.471 = 0.0316.$$

Take 3/2 power:

$$(0.0316)^{3/2} = 0.0316 \cdot \sqrt{0.0316} = 0.0316 \cdot 0.1778 = 5.62 \times 10^{-3}.$$

Compute Boltzmann exponent:

$$\beta E_g = \frac{1.42 \text{ eV}}{0.0259 \text{ eV}} = 54.93.$$

Exponentiate:

$$e^{-54.93} = 1.41 \times 10^{-24}.$$

Combine:

$$n_i^2(300 \text{ K}) = (2.51 \times 10^{25})^2 \cdot 5.62 \times 10^{-3} \cdot 1.41 \times 10^{-24}.$$

Square the prefactor:

$$(2.51 \times 10^{25})^2 = 6.30 \times 10^{50}.$$

Multiply numerical factors:

$$n_i^2(300 \text{ K}) = 6.30 \times 10^{50} \cdot 5.62 \times 10^{-3} \cdot 1.41 \times 10^{-24} = 4.99 \times 10^{24} \text{ m}^{-6}.$$

Square-root:

$$n_i(300 \text{ K}) = 2.23 \times 10^{12} \text{ m}^{-3}.$$

At $T = 150 \text{ K}$ scale: $N_0 \propto T^{3/2}$, so $N_0^2 \propto T^3$; βE_g doubles.

$$N_0^2(150 \text{ K}) = N_0^2(300 \text{ K}) \cdot (1/2)^3 = 0.125 \cdot 6.30 \times 10^{50}.$$

Boltzmann exponent at 150 K:

$$\beta E_g = 2 \cdot 54.93 = 109.86.$$

Exponentiate (square of $e^{-54.93}$):

$$e^{-109.86} = (1.41 \times 10^{-24})^2 = 1.99 \times 10^{-48}.$$

Multiply:

$$n_i^2(150 \text{ K}) = 0.125 \cdot 6.30 \times 10^{50} \cdot 5.62 \times 10^{-3} \cdot 1.99 \times 10^{-48}.$$

Reduce:

$$n_i^2(150 \text{ K}) = 0.88 \text{ m}^{-6}.$$

Square-root:

$$n_i(150 \text{ K}) \approx 0.94 \text{ m}^{-3} \quad (\text{essentially zero}).$$

p-doped GaAs, $N_A^- = 10^{11} \text{ m}^{-3}$ fully ionised. Charge neutrality:

$$p = n + N_A^-.$$

Combine with (6): $n(n + N_A^-) = n_i^2$.

Solve the quadratic for n :

$$n = \frac{1}{2} \left[-N_A^- + \sqrt{(N_A^-)^2 + 4n_i^2} \right].$$

At 300 K: $n_i = 2.23 \times 10^{12} \text{ m}^{-3} \gg N_A^-$, near-intrinsic regime.

Evaluate the discriminant:

$$(10^{11})^2 + 4 \cdot 4.99 \times 10^{24} = 10^{22} + 1.996 \times 10^{25} \approx 2.0 \times 10^{25}.$$

Take square-root:

$$\sqrt{2.0 \times 10^{25}} = 4.47 \times 10^{12} \text{ m}^{-3}.$$

Subtract N_A^- and halve:

$$n = \frac{1}{2}(-10^{11} + 4.47 \times 10^{12}) = \frac{1}{2} \cdot 4.37 \times 10^{12} = 2.18 \times 10^{12} \text{ m}^{-3}.$$

Add N_A^- for p :

$$p = n + N_A^- = 2.18 \times 10^{12} + 10^{11} = 2.28 \times 10^{12} \text{ m}^{-3}.$$

At 150 K: $n_i^2 \ll (N_A^-)^2$, fully extrinsic.

$$p \approx N_A^- = 10^{11} \text{ m}^{-3}, \quad n = n_i^2/p \approx 0.88/10^{11} \approx 8.8 \times 10^{-12} \text{ m}^{-3} \text{ (i.e. zero)}.$$

T	n	p
300 K	$2.18 \times 10^{12} \text{ m}^{-3}$	$2.28 \times 10^{12} \text{ m}^{-3}$
150 K	≈ 0	10^{11} m^{-3}

(e) Experiment

Hall effect. Pass current I along $\hat{x}$ through a thin slab in field $B\hat{z}$, measure transverse Hall voltage V_H . The Hall coefficient (two-carrier formula) is

$$R_H = \frac{V_H t}{IB} = \frac{1}{e} \frac{p\mu_h^2 - n\mu_e^2}{(p\mu_h + n\mu_e)^2}.$$

Combine with the conductivity $\sigma = e(n\mu_e + p\mu_h)$, and measure $R_H(T)$ and $\sigma(T)$ to extract $n(T)$, $p(T)$ separately.

Sensitivity. R_H weights each carrier by μ^2 . In GaAs $\mu_e \approx 8500 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1} \gg \mu_h \approx 400 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, so the measurement is dominated by **electrons** until the hole density exceeds $(\mu_e/\mu_h)^2 \approx 450$ times the electron density.

Question 4 — 1D ferromagnet domain wall

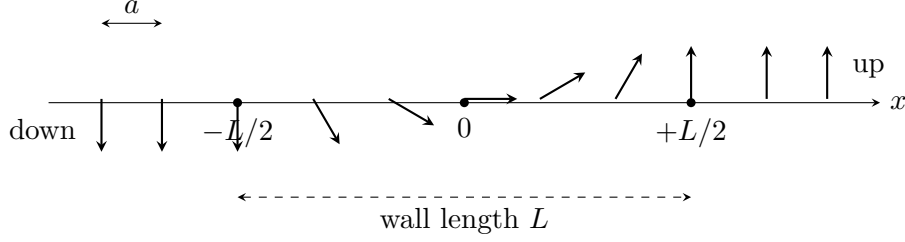
Problem. (a) Discuss the Hamiltonian terms, sketch the chain, justify ferromagnetic alignment far from the wall. (b) Domain-wall energy as a function of L . (c) Optimal L and minimum energy. (d) 2D wall energy and energy/volume of a stripe array. (e) Equilibrium $D \propto t^\beta$, find β .

(a) Hamiltonian and chain sketch

Exchange term $-J \sum \mathbf{S}_i \cdot \mathbf{S}_j$ ($J > 0$): nearest-neighbour ferromagnetic coupling, minimised when neighbouring spins are parallel.

Anisotropy term $-\kappa \sum (S_i^z)^2$ ($\kappa > 0$): single-ion uniaxial easy-axis term, minimised when each spin lies along $\pm \hat{z}$.

Far from the wall both terms are simultaneously minimised by $\mathbf{S}_i = +S\hat{z}$ (or $-S\hat{z}$) on every site, so the chain is uniformly ferromagnetic at $|x| \gg L$.



(b) Wall energy as a function of L

Spins rotate uniformly across the wall: take angle from $\hat{z}$ at site i to be $\theta_i = \pi/2 - \pi x_i/L$. Project onto $\hat{z}$:

$$S_i^z = S \cos \theta_i = S \cos(\pi/2 - \pi x_i/L) = S \sin(\pi x_i/L).$$

Inter-site rotation:

$$\delta\theta = \theta_{i+1} - \theta_i = -\pi a/L, \quad |\delta\theta| = \pi a/L.$$

Exchange contribution. Each bond contributes $-JS^2 \cos \delta\theta$. Subtract the uniform-FM bond energy $-JS^2$:

$$\Delta E_{\text{ex,bond}} = -JS^2 \cos \delta\theta - (-JS^2) = JS^2(1 - \cos \delta\theta).$$

Small-angle expand (since $L \gg a$):

$$1 - \cos \delta\theta \approx \frac{1}{2}(\delta\theta)^2.$$

Insert $\delta\theta = \pi a/L$:

$$\Delta E_{\text{ex,bond}} \approx \frac{1}{2}JS^2 \left(\frac{\pi a}{L}\right)^2 = \frac{\pi^2 JS^2 a^2}{2L^2}.$$

Sum over L/a bonds in the wall:

$$\Delta E_{\text{ex}} = \frac{L}{a} \cdot \frac{\pi^2 JS^2 a^2}{2L^2} = \frac{\pi^2 JS^2 a}{2L}.$$

Anisotropy contribution. Site energy is $-\kappa(S_i^z)^2 = -\kappa S^2 \sin^2(\pi x_i/L)$. Subtract uniform $-\kappa S^2$:

$$\Delta E_{\text{an,site}} = -\kappa S^2 \sin^2(\pi x_i/L) + \kappa S^2 = \kappa S^2(1 - \sin^2(\pi x_i/L)).$$

Apply $\sin^2 + \cos^2 = 1$:

$$\Delta E_{\text{an,site}} = \kappa S^2 \cos^2(\pi x_i/L).$$

Convert sum to integral via the hint, $\sum_m f(m\delta) \rightarrow (1/\delta) \int f(x) dx$ with $\delta = a$:

$$\Delta E_{\text{an}} = \frac{1}{a} \int_{-L/2}^{L/2} \kappa S^2 \cos^2(\pi x/L) dx.$$

Use $\int_{-L/2}^{L/2} \cos^2(\pi x/L) dx = L/2$:

$$\Delta E_{\text{an}} = \frac{1}{a} \cdot \kappa S^2 \cdot \frac{L}{2} = \frac{\kappa S^2 L}{2a}.$$

Total wall energy:

$$\boxed{\Delta E(L) = \frac{\pi^2 JS^2 a}{2L} + \frac{\kappa S^2 L}{2a}}. \quad (7)$$

(c) Minimisation

Differentiate (7) with respect to L :

$$\frac{d\Delta E}{dL} = -\frac{\pi^2 JS^2 a}{2L^2} + \frac{\kappa S^2}{2a}.$$

Set to zero:

$$\frac{\kappa S^2}{2a} = \frac{\pi^2 JS^2 a}{2L^2}.$$

Cancel $S^2/2$ on each side:

$$\frac{\kappa}{a} = \frac{\pi^2 J a}{L^2}.$$

Solve for L^2 :

$$L^2 = \frac{\pi^2 J a^2}{\kappa}.$$

Take square-root:

$$L = \pi a \sqrt{J/\kappa}, \quad c_1 = \pi.$$

Substitute back into the anisotropy term:

$$\frac{\kappa S^2 L}{2a} = \frac{\kappa S^2}{2a} \cdot \pi a \sqrt{J/\kappa} = \frac{\pi S^2}{2} \cdot \kappa \sqrt{J/\kappa} = \frac{\pi S^2}{2} \sqrt{J\kappa}.$$

By the symmetry of the optimisation the exchange term equals the anisotropy term:

$$\Delta E_{\min} = 2 \cdot \frac{\pi S^2}{2} \sqrt{J\kappa} = \pi S^2 \sqrt{J\kappa}.$$

Box:

$$\Delta E_{\min} = \pi S^2 \sqrt{J\kappa}.$$

(d) 2D wall and stripe array

(i) Single wall. The 2D lattice has Wt/a^2 parallel chains threading the y - z cross-section, each contributing $\Delta E_{\min}$:

$$E_{\text{wall}} = \frac{Wt}{a^2} \pi S^2 \sqrt{J\kappa}.$$

(ii) Stripe array. Walls are spaced D along x ; the slab volume associated with one wall is $D \cdot W \cdot t$.

$$\frac{E_{\text{walls}}}{V_{\text{per wall}}} = \frac{(Wt/a^2) \pi S^2 \sqrt{J\kappa}}{D W t}.$$

Cancel Wt :

$$\frac{E_{\text{walls}}}{V} = \frac{\pi S^2 \sqrt{J\kappa}}{a^2 D}. \quad (8)$$

(e) Equilibrium domain spacing

Domain walls cost positive exchange + anisotropy energy, but breaking the magnet into alternating-direction stripes lowers the long-range **magnetostatic (stray-field) energy**, since the surface poles partially cancel between adjacent stripes. Walls form when the magnetostatic saving exceeds the wall cost.

Total energy per unit volume (sum (8) and the given magnetostatic term):

$$\frac{E}{V} = \frac{\pi S^2 \sqrt{J\kappa}}{a^2 D} + c_2 \frac{\mu_0 M^2 D}{t}.$$

Differentiate with respect to D at fixed t :

$$\frac{d}{dD} \left(\frac{E}{V} \right) = -\frac{\pi S^2 \sqrt{J\kappa}}{a^2 D^2} + \frac{c_2 \mu_0 M^2}{t}.$$

Set to zero:

$$\frac{c_2 \mu_0 M^2}{t} = \frac{\pi S^2 \sqrt{J\kappa}}{a^2 D^2}.$$

Rearrange for D^2 :

$$D^2 = \frac{\pi S^2 \sqrt{J\kappa} t}{c_2 \mu_0 M^2 a^2}.$$

Take square-root, so $D \propto \sqrt{t}$:

$D \propto t^{1/2}, \quad \beta = \frac{1}{2}.$
